“A Statistical Analysis of COVID-19 Trends Using Time Series Data”

A PROJECT SUBMITTED TO

SYNTECXHUB Technologies

Kanpur Nagar, Uttar Pradesh

FOR THE INTERNSHIP

IN

DATA SCIENCE

By

Gadhave Swati Sanjay

22-04-2026

Introduction

The outbreak of Coronavirus Disease 2019 (COVID-19) emerged as a global health crisis, significantly impacting societies, economies, and healthcare systems worldwide. Since its identification in late 2019, the pandemic has spread rapidly across countries, leading to unprecedented challenges in monitoring, controlling, and understanding its progression. In such a scenario, data-driven analysis plays a crucial role in assessing the spread and severity of the disease.

Time-series data of COVID-19 cases provides valuable insights into how the pandemic has evolved over time. By analyzing confirmed case counts across different countries, it becomes possible to identify trends, detect patterns of growth, and understand variations in infection rates. Such analysis is essential for evaluating the effectiveness of public health interventions and for comparing the progression of the pandemic across regions.

This project focuses on the analysis of COVID-19 confirmed case data for selected countries, namely India, the United States, Brazil, and the United Kingdom. Using statistical and visualization techniques, the study aims to examine cumulative trends, daily case fluctuations, and smoothed patterns through rolling averages. Additionally, the analysis identifies peak infection periods and explores weekly trends to provide a comprehensive understanding of the pandemic's behaviour.

Overall, this study demonstrates how data analysis techniques can be effectively applied to real-world datasets to extract meaningful insights, supporting informed decision-making and enhancing our understanding of large-scale public health events.

Objectives:

- To analyze the overall trend of COVID-19 confirmed cases across selected countries
- To examine daily new COVID-19 cases and identify patterns of spread
- To smooth daily fluctuations using a 7-day rolling average to better understand trends
- To identify the peak period of COVID-19 cases for each country
- To analyze weekly trends in COVID-19 cases

Methodology:

Data Source:
The dataset used for this project is the COVID-19 Time Series Dataset, obtained from the Johns Hopkins University Center for Systems Science and Engineering (JHU CSSE). The dataset is publicly available on GitHub and is widely used by researchers and analysts for tracking and analyzing the spread of COVID-19 across the world.
It contains country-wise daily records of confirmed COVID-19 cases in a time-series format, where each column represents a specific date and each row corresponds to a country or region.

Variables Description:

- **Country/Region:** Name of the country
- **Province/State:** State or region within the country (if applicable)
- **Date:** Specific date of observation
- **Confirmed:** Total cumulative confirmed COVID-19 cases
- **Daily Cases:** Number of new cases reported per day (derived variable)
- **Rolling Average:** 7-day average of daily cases (derived variable)

Process:
First, the dataset was loaded using the Pandas library and basic exploratory analysis was performed to understand its structure. The data was then converted from wide format to long format, and the date column was transformed into datetime format.

Next, the data was aggregated at the country level, and selected countries were filtered for analysis. New variables such as daily cases and a 7-day rolling average were created to analyze trends.

Finally, graphs were plotted to study total cases, daily variations, and smoothed trends, and peak values were identified to draw meaningful conclusions.

Exploratory Data Analysis (EDA) & Data Preparation

1. **Data Loading**

Code:

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

df=pd.read_csv("C:\\Users\\ACER\\Downloads\\time_series_covid19_confirmed_global.csv"
)
```

2. **Missing Values Handling**

Code:

```
df.isnull().sum()
```

Result:

Missing values are mainly in the Province/State column and do not affect the analysis, as it is conducted at the country level.

3. **Data Transformation (Wide to Long Format)**

Code:

```
df_long = df.melt(

    id_vars=['Province/State', 'Country/Region', 'Lat', 'Long'],

    var_name='Date',

    value_name='Confirmed'

)

df_long.head()
```

Output:

	Province/State	Country/Region	Lat	Long	Date	Confirmed
0	NaN	Afghanistan	33.93911	67.709953	1/22/20	0
1	NaN	Albania	41.15330	20.168300	1/22/20	0
2	NaN	Algeria	28.03390	1.659600	1/22/20	0
3	NaN	Andorra	42.50630	1.521800	1/22/20	0
4	NaN	Angola	-11.20270	17.873900	1/22/20	0

4. **Date Conversion**

Code:

```
df_long['Date'] = pd.to_datetime(df_long['Date'], format='%m/%d/%y', errors='coerce')
df_long.head()
```

Output:

	Province/State	Country/Region	Lat	Long	Date	Confirmed
0	NaN	Afghanistan	33.93911	67.709953	2020-01-22	0
1	NaN	Albania	41.15330	20.168300	2020-01-22	0
2	NaN	Algeria	28.03390	1.659600	2020-01-22	0
3	NaN	Andorra	42.50630	1.521800	2020-01-22	0
4	NaN	Angola	-11.20270	17.873900	2020-01-22	0

5. **Data Aggregation**

Code:

```
country_df = df_long.groupby(['Country/Region', 'Date'])['Confirmed'].sum().reset_index()
country_df.head()
```

Output:

	Country/Region	Date	Confirmed
0	Afghanistan	2020-01-22	0
1	Afghanistan	2020-01-23	0
2	Afghanistan	2020-01-24	0
3	Afghanistan	2020-01-25	0
4	Afghanistan	2020-01-26	0

6. **Data Filtering (Selected Countries)**

Code:

```
countries = ['India', 'US', 'Brazil', 'United Kingdom']
data = country_df[country_df['Country/Region'].isin(countries)].copy()
data.head()
```

Output:

	Country/Region	Date	Confirmed
16560	Brazil	2020-01-22	0
16561	Brazil	2020-01-23	0
16562	Brazil	2020-01-24	0
16563	Brazil	2020-01-25	0
16564	Brazil	2020-01-26	0

7. **Feature Engineering (Daily Cases Calculation)**

Code:

```
data['Daily Cases'] = data.groupby('Country/Region')['Confirmed'].diff().fillna(0)

data.head()
```

Output:

	Country/Region	Date	Confirmed	Daily Cases
16560	Brazil	2020-01-22	0	0.0
16561	Brazil	2020-01-23	0	0.0
16562	Brazil	2020-01-24	0	0.0
16563	Brazil	2020-01-25	0	0.0
16564	Brazil	2020-01-26	0	0.0

Data Analysis

- **To analyze the overall trend of COVID-19 confirmed cases across selected countries**

Code:

```
plt.figure(figsize=(12,6))

for country in countries:
    subset = data[data['Country/Region'] == country]
    plt.plot(subset['Date'], subset['Confirmed'], label=country)

plt.title("Total COVID-19 Cases Over Time")
plt.xlabel("Date")
plt.ylabel("Confirmed Cases")
plt.legend()

plt.xticks(rotation=45)
plt.tight_layout()

plt.show()
```

Output:

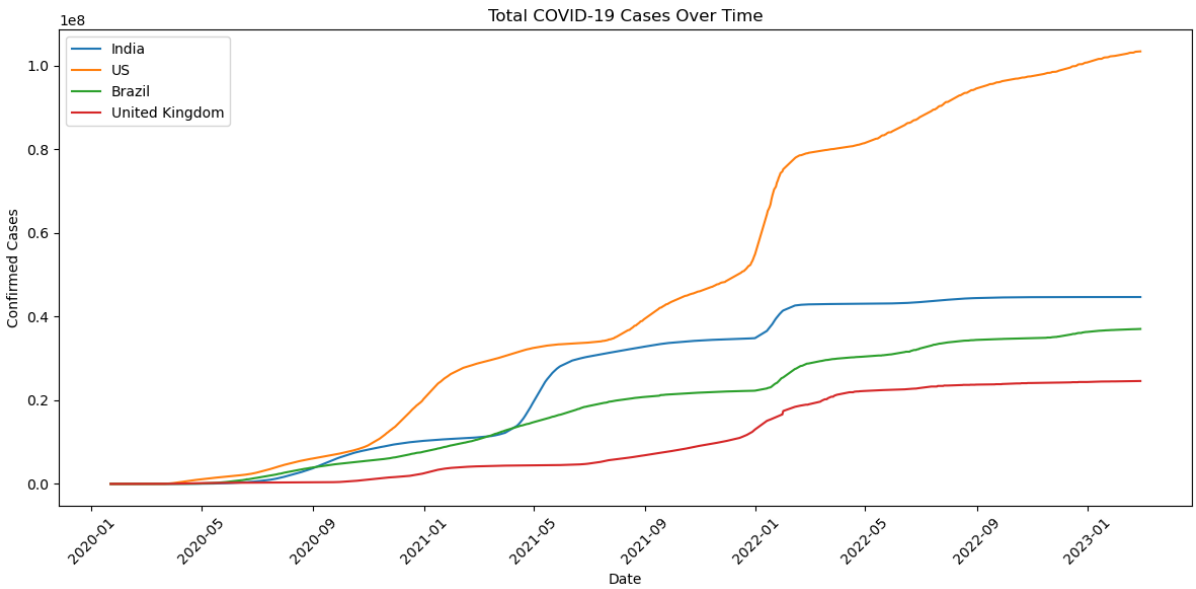

Figure 1: Total COVID-19 Cases Over Time

Interpretation:

The graph shows a continuous upward trend in confirmed COVID-19 cases for all countries. The United States exhibits the highest number of total cases, followed by India and Brazil. The steady increase indicates the cumulative nature of the dataset and reflects the global spread of the pandemic over time.

Conclusion:

COVID-19 cases increased consistently across all selected countries, with significant variation in magnitude, indicating differences in spread and population impact.

- **To examine daily new COVID-19 cases and identify patterns of spread**

Code:

```
plt.figure(figsize=(12,6))

for country in countries:
    subset = data[data['Country/Region'] == country]
    plt.plot(subset['Date'], subset['Daily Cases'], label=country)

plt.title("Daily COVID-19 Cases")
plt.xlabel("Date")
plt.ylabel("Daily Cases")
plt.legend()

plt.xticks(rotation=45)
plt.tight_layout()

plt.show()
```

Output:

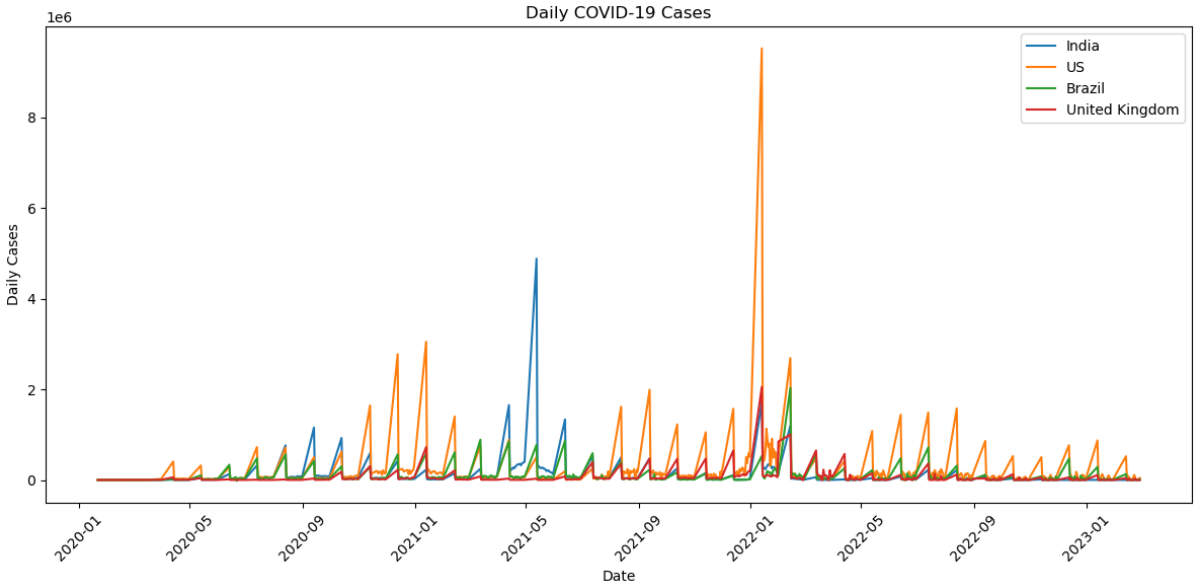

Figure 2: Daily COVID-19 Cases

Interpretation:

The daily cases graph shows fluctuations with several sharp spikes, indicating sudden increases in infection rates. These spikes represent different waves of the pandemic. The variability suggests that COVID-19 transmission was not constant but occurred in bursts over time.

Conclusion:

COVID-19 spread occurred in waves, with periods of rapid increase followed by stabilization, highlighting the dynamic nature of the pandemic.

- **To smooth daily fluctuations using a 7-day rolling average to better understand trends**

Code:

```
data['Rolling Avg'] = data.groupby('Country/Region')['Daily Cases']\
                    .transform(lambda x: x.rolling(7).mean())

plt.figure(figsize=(12,6))

for country in countries:
    subset = data[data['Country/Region'] == country]
    plt.plot(subset['Date'], subset['Rolling Avg'], label=country)

plt.title("7-Day Rolling Average of COVID-19 Cases")
plt.xlabel("Date")
plt.ylabel("Cases (Smoothed)")
plt.legend()

plt.xticks(rotation=45)
plt.tight_layout()

plt.show()
```

Output:

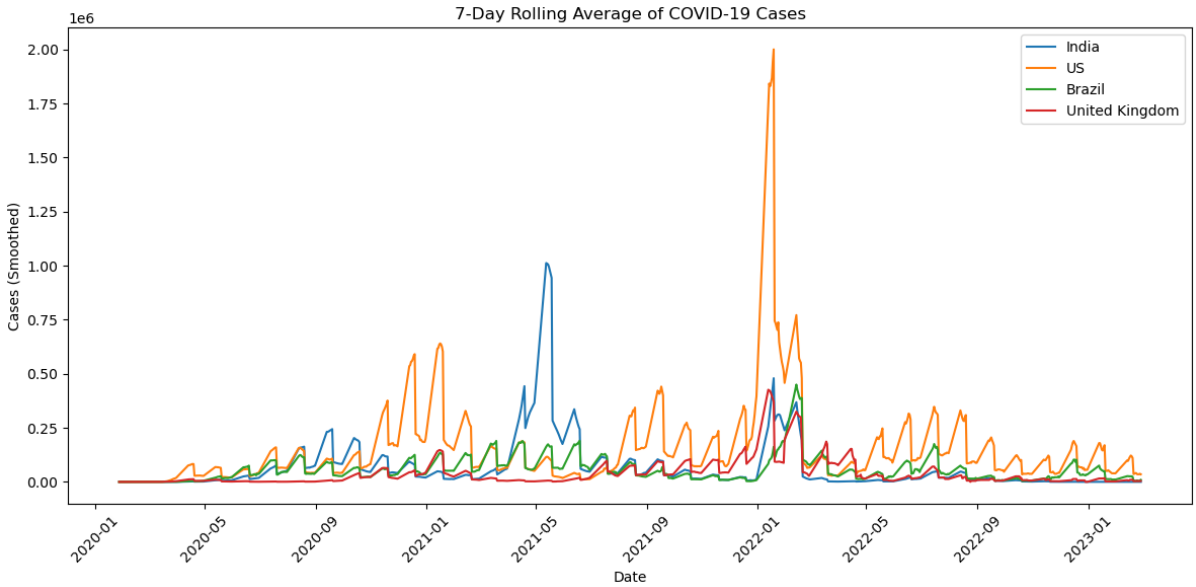

Figure 3: 7-Day Rolling Average of Cases

Interpretation:

The rolling average graph reduces noise and clearly highlights major waves of COVID-19 infections. It shows distinct peaks corresponding to different phases of the pandemic, making it easier to compare trends across countries.

Conclusion:

The 7-day rolling average provides a clearer view of infection trends, helping to identify major waves and compare the severity of outbreaks across countries.

- **To identify the peak period of COVID-19 cases for each country**

Code:

```
peaks = data.loc[data.groupby('Country/Region')['Rolling Avg'].idxmax()]

peaks = peaks.reset_index(drop=True)

peaks['Rolling Avg'] = peaks['Rolling Avg'].round(0).astype(int)

print(peaks[['Country/Region', 'Date', 'Rolling Avg']])
```

Output:

	Country/Region	Date	Rolling Avg
0	Brazil	2022-02-13	450083
1	India	2021-05-13	1012377
2	US	2022-01-19	2000844
3	United Kingdom	2022-01-13	426211

Interpretation:

The results show that each country experienced its peak COVID-19 cases at different times. India reached its peak in May 2021, while the US, Brazil, and the United Kingdom peaked in early 2022. The United States recorded the highest peak, followed by India, indicating differences in the severity and timing of the pandemic across countries.

Conclusion:

The peak analysis highlights that COVID-19 waves occurred at different times and intensities across countries, reflecting variations in the spread and progression of the pandemic.

- **To analyze weekly trends in COVID-19 cases**

Code:

```
data['Week'] = data['Date'].dt.to_period('W')

weekly_filtered = (

    data.groupby(['Country/Region', 'Week'])['Daily Cases']

        .sum()

        .reset_index()

)

weekly_filtered = weekly_filtered[weekly_filtered['Daily Cases'] > 0]\

                                .reset_index(drop=True)

weekly_filtered['Daily Cases'] = weekly_filtered['Daily Cases'].astype(int)

weekly_filtered.head()
```

Output:

	Country/Region	Week	Daily Cases
0	Brazil	2020-02-24/2020-03-01	2
1	Brazil	2020-03-09/2020-03-15	160
2	Brazil	2020-03-16/2020-03-22	1384
3	Brazil	2020-03-23/2020-03-29	2710
4	Brazil	2020-03-30/2020-04-05	1461

Interpretation:

The weekly data shows how COVID-19 cases increased over time. In the beginning, there were very few or no cases, but gradually the number of cases started rising. As weeks passed, the increase became faster, showing how the virus spread from a small outbreak to a larger scale. Looking at data week by week makes it easier to understand the overall pattern without getting confused by daily ups and downs.

Conclusion:

By grouping the data into weeks, it becomes easier to see how COVID-19 spread over time. It clearly shows the shift from early low cases to a rapid increase, helping us understand the overall growth of the pandemic in a simple way.

Overall CONCLUSION:

The analysis of COVID-19 time-series data reveals that the pandemic followed different patterns across countries in terms of growth, intensity, and timing. All selected countries showed a steady increase in cumulative cases, reflecting the widespread nature of the outbreak. However, daily case trends highlighted that the spread occurred in multiple waves, with periods of rapid increase followed by stabilization.

The use of a 7-day rolling average helped in clearly identifying major trends by reducing daily fluctuations, making it easier to compare the severity of outbreaks across countries. Peak analysis showed that each country experienced its highest number of cases at different times, indicating variations in the progression of the pandemic and response measures.

Weekly analysis further simplified the understanding of the spread by showing a gradual shift from low initial cases to rapid growth. Overall, the study demonstrates that time-series analysis is an effective tool for understanding pandemic trends and provides valuable insights into the dynamics of COVID-19 across different regions.